B04 Team1 EG1311 Report

Students: Tan Ping Hui,

1. Introduction

The aim of the robot is to be able to travel over a 3-cm tall bump and a 10-cm tall ramp, throw a ping-pong ball over a 30-cm wall while stopping within 5 cm before the wall, and then move back to the original point. The development of the robot involved balancing weight, stability, mobility and ball-handling requirements within the permitted dimensions. This report discusses the main difficulties encountered, the modifications made and the lessons learnt from developing the wheels, electrical connections and ping-pong ball mechanisms.

2. Wheel Design and Development

2.1. Wheel Diameter and Ramp Clearance

The initial design prioritised minimising the robot's weight. Wheels with 8 cm diameter were designed and laser-cut on the assumption that they are sufficient to cross the 3-cm tall bump while reducing material use. However, testing showed that the robot could not even clear the bump with the wheel dimensions. The wheel was therefore redesigned with a larger diameter of 10cm. After the change, it was able to overcome the bump, though inconsistently, as it seems to slip repeatedly due to the lack of traction.

[insert photo]

Figure X. Increased wheel diameter

This revision demonstrated the limitations of selecting dimensions based solely on an estimate. Although reducing weight was an important objective, the eventual goal is for the robot to cross the obstacles and secure the points. The experience highlighted the importance of verifying feasible dimensions through physical testing before finalising the design.

2.2 Wheel Cut-Outs and Structural Stability

To reduce wheel weight while retaining structural stability, we took ideas from the wheel design of modern cars. For the wheels of our robot, three cut-outs were incorporated into each circular wheel. These openings removed material while preserving the outer rim and the supporting sections connecting it to the central hub.

	
Fig X. Wheel of a Modern Car	Fig. X. Wheel of our robot

2.3 Wheel Friction

As aforementioned, during the test runs, our robot slips repeatedly while climbing over the bump despite adjusting the wheel diameter. This is because the smooth wheel surfaces provided insufficient grip during ramp climbing. Therefore, we cut the friction pads into strips and glued the strips around the circumference of the wheels. It is to increase traction between the wheels and the ramp. Afterwhich, the robot manage to climb up the bump consistently without any issues.

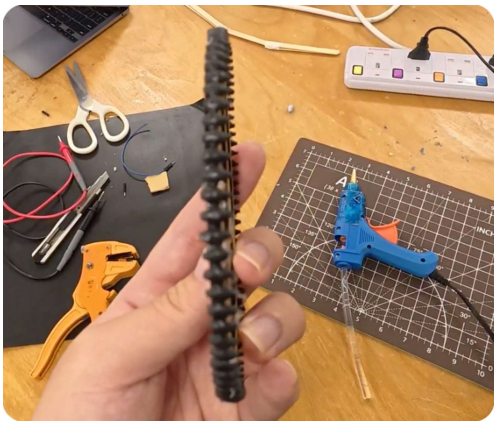


Figure X. Friction pads attached to the wheel surfaces.

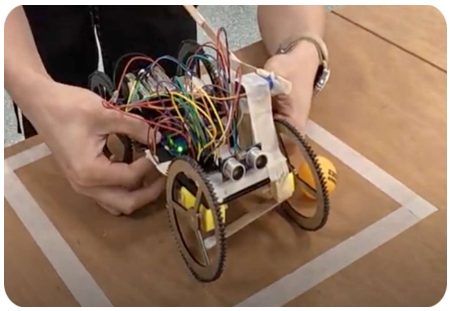
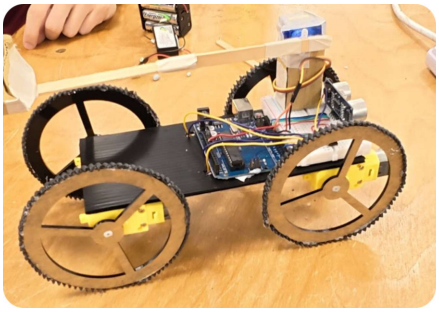
This modification addressed a different aspect of mobility from wheel diameter. While wheel size affected clearance, the contact surface influenced the robot's ability to transmit driving force without slipping. The experience demonstrated that climbing performance depends on several interacting factors. Both component dimensions and material properties must therefore be considered when designing a robot for movement over obstacles.

3. Electrical Wiring and Reliability

Electrical wiring was one of the most significant difficulties encountered during development. Some wires were damaged internally, meaning that their condition could not be determined through visual inspection alone. Individual wires had to be tested using a multimeter to identify faulty connections.

Connections also became loose during test runs. As a result, motor operation was inconsistent: at times, only two of the four motors operated, while on other occasions none operated. These intermittent faults complicated troubleshooting because the robot's behaviour changed between tests.

In order to solve the issue, the wiring arrangement was simplified to reduce unnecessary wire length and improve organisation. Shorter wires were used where practical, while sufficient length was retained to avoid tension at the connections. Connections were secured, and some longer wires were routed beneath the base. This approach minimises the long wires getting trapped with the wheels during runs, and it is easier for us to track and troubleshoot the faults.

	
Fig X. Original Wiring	Fig. X. Final Wiring

The main lesson from this process was that reliable electrical connections are essential to consistent robot performance. Mechanical components and control code cannot function as intended if power or control connections are interrupted, hindering testing and extending development time unnecessarily. However, testing wires individually also demonstrated to us the value of systematic fault isolation. We need to find a balance between quick prototyping with jumper cables and reliable implementation using crimmed wires. A more organised and accessible wiring layout supports both reliable operation and easier maintenance.

4. Ping-Pong Ball Mechanism

4.1 Ball Holder Arm Length and Mounting Position

The length of the ping-pong ball holder had to satisfy two requirements: it needed to remain within the robot's permitted dimensions and allow the launching mechanism to throw the ball over a 30 cm wall. As the arm rested horizontally, its length had to be kept below 22 cm after accounting for the ball box dimensions and mounting position. This restricted the height that the ball box could reach when the servo rotated the arm.

We first mounted the mechanism at the far end of the robot's base to maximise the arm reach within the available space. However, this arrangement still did not provide sufficient height for the ball to clear the wall. Increasing the arm length further would have exceeded the dimensional constraint, so we considered raising the entire mechanism instead.

Since the robot was still well below the 30 cm height limit, we used a cardboard box to elevate the servo and ball holder arm. We calculated the required support height so that the ball box would rise just above the wall during the launching motion. This allowed us to use the available vertical space while keeping the arm within the horizontal dimensional limit.

[insert picture here]

Figure 5. Ping-pong ball holder and its position within the robot.

The support height also involved a trade-off. Raising the mechanism increased its reach, but excessive elevation would raise the robot's centre of gravity and could make it less stable when accelerating or crossing obstacles. We therefore selected a height that provided the required clearance without unnecessarily raising the mechanism. After these changes, the ball finally managed to clear the wall.

This process taught us to consider arm length, mounting position and mounting height together. When one dimension was constrained, adjusting another offered a practical solution. It also showed that improving launch clearance had to be balanced against the stability of the complete robot.

4.2 Ball Box Dimensions and Retention

The connection between the arm and ball box also required careful consideration. We mounted the arm at the centre of the box's base rather than along one side to distribute the load more evenly about the connection. Two ice cream sticks, each with a notch cut into one end, were fitted together to form an interlocking support.

The ball box needed to retain the ball while the robot travelled over the bump and ramp, while still allowing it to leave during launching. Its dimensions therefore had to balance containment and release. A box that held the ball securely when stationary might still allow it to escape during acceleration or obstacle crossing, so checking its fit alone was insufficient.

These considerations highlighted the importance of assessing the complete mechanism under its intended operating conditions. The arm, mounting joint and ball box had to work together throughout both travel and launching, rather than simply fit together in the stationary robot.

[insert picture here]

Figure 5. Ball holder positioned at the centre of the box's base using two interlocking ice cream sticks.

5. Graded Runs and Final Amendment

After securing points during the initial graded runs, our group focused on reducing the robot's weight and refining its component layout as our robot was the second heaviest among the other groups. We removed unnecessary sections of the base and trimmed the excess length of the ice cream sticks used to join components. These modifications aimed to reduce material use while preserving the support required by the structure.

We also rearranged the battery placement, positioning the 5 V battery above the breadboard to make the layout more compact. In total, we reduced the weight by more than 10 grams, without extensively changing the design to preserve the integrity of the robot.

[insert picture here]

Figure 7. Revised robot showing the trimmed base, shortened connecting sticks and rearranged battery placement.

6. Conclusion

The main lesson learnt was that initial design assumptions must be verified through practical testing. Early component tests, followed by testing of the complete assembly, provide opportunities to identify problems and revise the design. Future development would benefit from verification of dimensions, more deliberate wiring organisation and consistent testing under the robot's intended operating conditions.